

- Give it a name - test and choose the m1-tiny flavour.
- Select the boot source as 'Boot from image' with the image name 'cirros', a very small image included in the installation.
- Once it is launched, associate the floating ip obtained above with this instance.

Now, you are ready to log in to the VM created in your local cloud. In a terminal window, type:

```
ssh cirros@172.24.4.229
```

You should be signed into the virtual machine without needing a password.

You can experiment with importing the Fedora VM image you used for the OpenStack VM and launching it in the cloud. Whether you succeed or not will depend on the resources available in the OpenStack VM.

Installing only the needed OpenStack services

You will have observed that OpenStack comes with a very wide range of services, some of which are not likely to be very useful for your experiments on the desktop, e.g., the additional networks and router created in the tests above. Here is a part of the dialogue for installing the required services on the desktop:

```
[root@amd ~]# packstack
Welcome to Installer setup utility
Enter the path to your ssh Public key to install on servers:
Packstack changed given value to required value /root/.ssh/id_rsa.pub
Should Packstack install MySQL DB [y/n] [y] : y
Should Packstack install OpenStack Image Service (Glance) [y/n] [y] : y
Should Packstack install OpenStack Block Storage (Cinder) service [y/n] [y] : n
Should Packstack install OpenStack Compute (Nova) service [y/n] [y] : y
Should Packstack install OpenStack Networking (Neutron) service [y/n] [y] : n
Should Packstack install OpenStack Dashboard (Horizon) [y/n] [y] : y
Should Packstack install OpenStack Object Storage (Swift) [y/n] [y] : n
Should Packstack install OpenStack Metering (Ceilometer) [y/n] [y] : n
Should Packstack install OpenStack Orchestration (Heat) [y/n] [n] : n
```

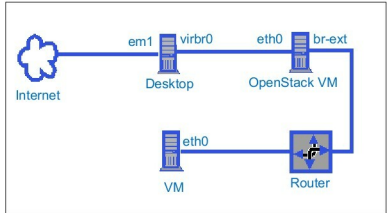


Figure 1: Simplified network diagram

```
Should Packstack install OpenStack client tools [y/n] [y] : y
```

The answers to the other questions will depend on the network interface and the IP address of your desktop, but there is no ambiguity here. You should answer with the interface 'lo' for `CONFIG_NOVA_COMPUTE_PRIVIF` and `CONFIG_NOVA_NETWORK_PRIVIF`. You don't need an extra physical interface as the compute services are running on the same server.

Now, you are ready to test your OpenStack installation on the desktop. You may want to create a project and add a user to the project. Under *Project/Compute/Access & Security*, you will need to add firewall rules and key pairs, as above.

However, you will not need to create any additional private network or a router.

Import a basic cloud image, e.g., from <http://fedoraproject.org/get-fedora#clouds> under *Project/Compute/Images*.

You may want to create an additional flavour for a virtual machine. The m1.tiny flavour has 512MB of RAM and 4GB of disk and is too small for running Hadoop. The m1.small flavour has 2GB of RAM and 20GB of disk, which will restrict the number of virtual machines you can run for testing Hadoop. Hence, you may create a mini flavour with 1GB of RAM and 10GB of disk. This will need to be done as the admin user.

Now, you can create an instance of the basic cloud image. The default user is `fedora` and your setup is ready for exploration of Hadoop data. **END** 

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